



ECE 639
Selected Topics in Electronics
Power Management
Spring 2021

Topic 1
Introduction

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(Some slides are courtesy of A. Fayed, Iowa State University and G. Rincon-Mora, Georgia-Tech)

Course Administration

- **Instructor:** Dr. Sameh Assem Ibrahim
- **e-mail address:** sameh.ibrahim@eng.asu.edu.eg
- **Reference Material:**
 - Gabriel Rincon-Mora, Power IC Design from the Ground Up, 3rd Edition, 2014
 - Lecture Notes
 - Selected Papers
- **Grading:**
 - Project: 50 %
 - Final: 50 %



Outline

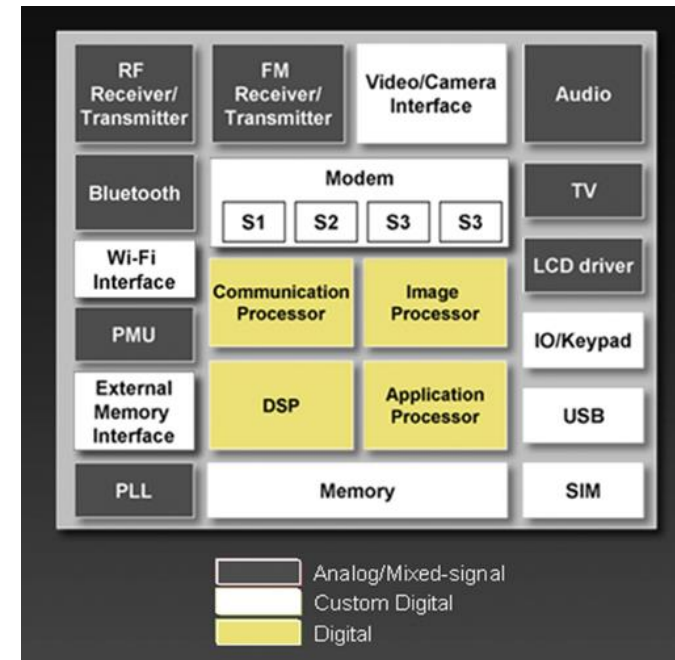
- **Introduction to Power Management**
- **Building Blocks of a PMU**
 - Bandgap Reference
 - LDOs
 - Buck
 - Battery Charging Controller
 - Energy Harvesters
- **Power Management Glossary**



The Road to SOCs

- **A typical system-on-chip SOC consists of:**

- A microcontroller, microprocessor or DSP core(s)
 - Taught in a digital design course.
- Memory blocks including a selection of ROM, RAM, EEPROM and flash memory
 - Taught in a digital design course.
- Timing sources including oscillators and phase-locked loops.
 - Taught in an analog system design course and in a serial-links design course.
- External interfaces including industry standards such as USB, FireWire, Ethernet, USART, SPI
 - Taught in a serial-links design course.
- Analog interfaces including ADCs and DACs
 - Taught in an analog system design course.
- Voltage regulators and power management circuits
 - Taught in this course.

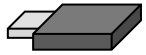


Power Management Perspectives

- A system designer → Task oriented (different states of operation)
- A digital designer → Dynamic voltage scaling
- A chip architect → Power delivery
- An analog/PM designer → Power conversion, regulation, and monitoring

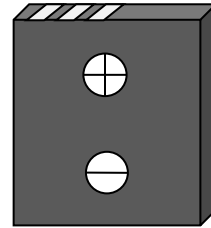


The Need for PMUs



Other Sources of Power

- Required for charging
- Needs conditioning



Battery



PMU

- Long lifetime
- Efficient charging/discharging
- Slow loss of charge
- High energy density



**Digital
Circuits**



**Analog
Circuits**

- Low supply for low power
- Fast transients affect other circuits
- Power proportional to supply
- Performance is important
- Different supplies

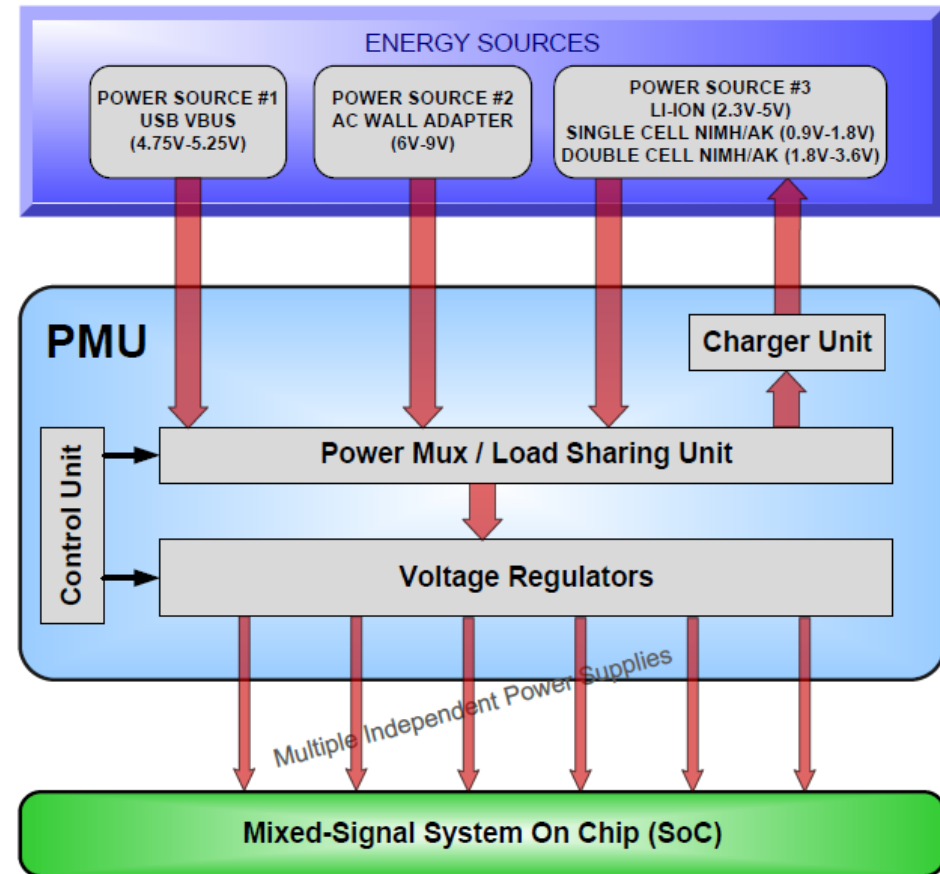
Power Management Unit

- **PMUs are used to manage the accurate power flow in portable and handheld devices, such as cell phone power amplifiers and LED display, CPU, DRAM, Graphics, High Speed I/O and USB.**
- **PMUs convert unpredictable and noisy supplies to stable, constant, accurate, and load-independent voltages.**
- **PMUs need to do these tasks with high efficiency to maximally extend battery life.**
- **PMUs are responsible for controlling the power flow from and into the battery.**
- **Minimize crosstalk between circuits sharing the same supply.**



Power Management Main Tasks

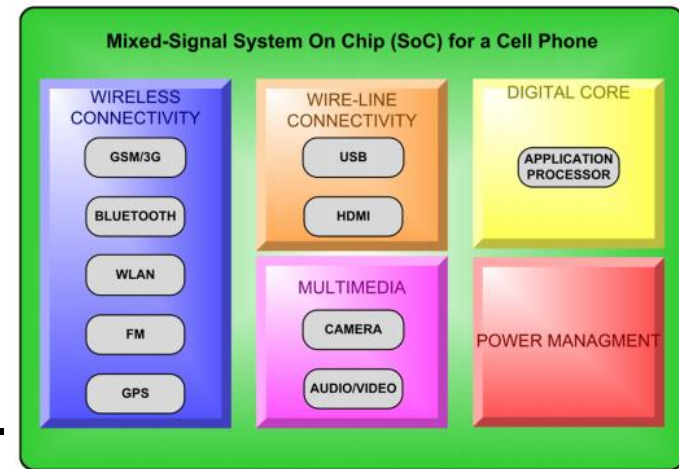
- Power source selection
- Load sharing
- Provide “regulated” power rails
- Supply level management
- Battery charging and gauging
- Leakage management



PM Nowadays Importance (1)

- **Move to SOC**

- PMUs have to be integrated onto the same die of the load adding design challenges.
- State-of-the-art microprocessors, digital-signal processing (DSP) circuits, and custom power intensive applications require PMUs to be quick, accurate, and capable of delivering high currents.
- Analog/Digital/RF is successfully fully integrated with minimal external components causing PMUs to be a real cost, size, and performance bottleneck.



PM Nowadays Importance (2)



- **Battery Life Challenge**

- Slow pace battery technology development
- Focus has been on power consumption reduction → technology scaling, low voltage/power design → diminishing returns?
- Thermal Efficiency
- The efficiency of how power is converted and delivered from the battery to the system must be revisited for any further improvement.



PM Nowadays Importance (3)



- **Going Green**

- Consumer electronics are becoming more ecology-conscious and consequently more energy-efficient and environment-friendly → Higher PMU efficiency are required.
- Energy harvesting is becoming mainstream.



PM New Challenges

- **Adaptability to different batteries or power sources**
- **Large number of independent, well-isolated power domains from a single shared source → often with drastically different requirements.**
- **Different types of loads → Analog, digital, RF**
- **With massive integration, there is no enough package pins to deliver all required power supply domains externally.**
- **Not easily portable/reusable from one SoC to another**



PMUs Boom



- All major IC companies are building their own PMU design groups.
- PMU design companies are increasing in number and gaining higher revenues.
- PMUs ICs market share is expected to reach \$35B by 2023.

Power Management Market Key Players

World Top 10 Power Semiconductor (US \$ Revenues)

2015 Rank	Manufacturer
1	Infineon Technologies*
2	Texas Instruments
3	STMicroelectronics
4	Maxim Integrated
5	Qualcomm
6	ON Semiconductor
7	NXP
8	Fairchild Semiconductor
9	Renesas Electronics Corporation
10	Linear Technology
	International Rectifier*
	Others

Total Market Size =

Notes

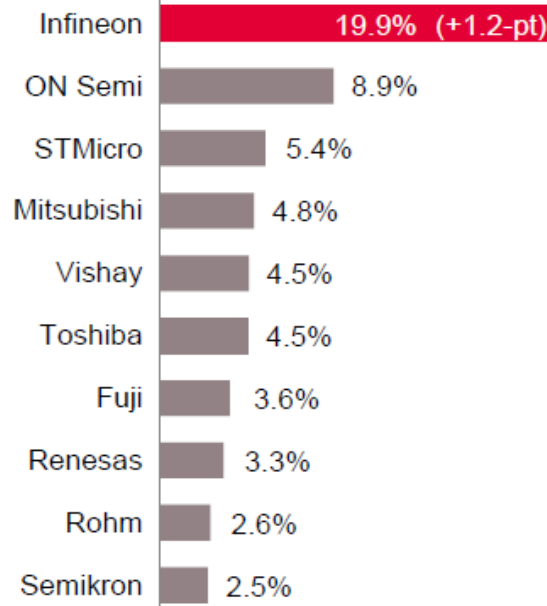
* Infineon Technologies acquired International Rectifier in 2015 and combined as Infineon in 2015.

r= revised since 2015 edition

Source: IHS

Power discretes and modules

total market in 2018: \$21.0bn



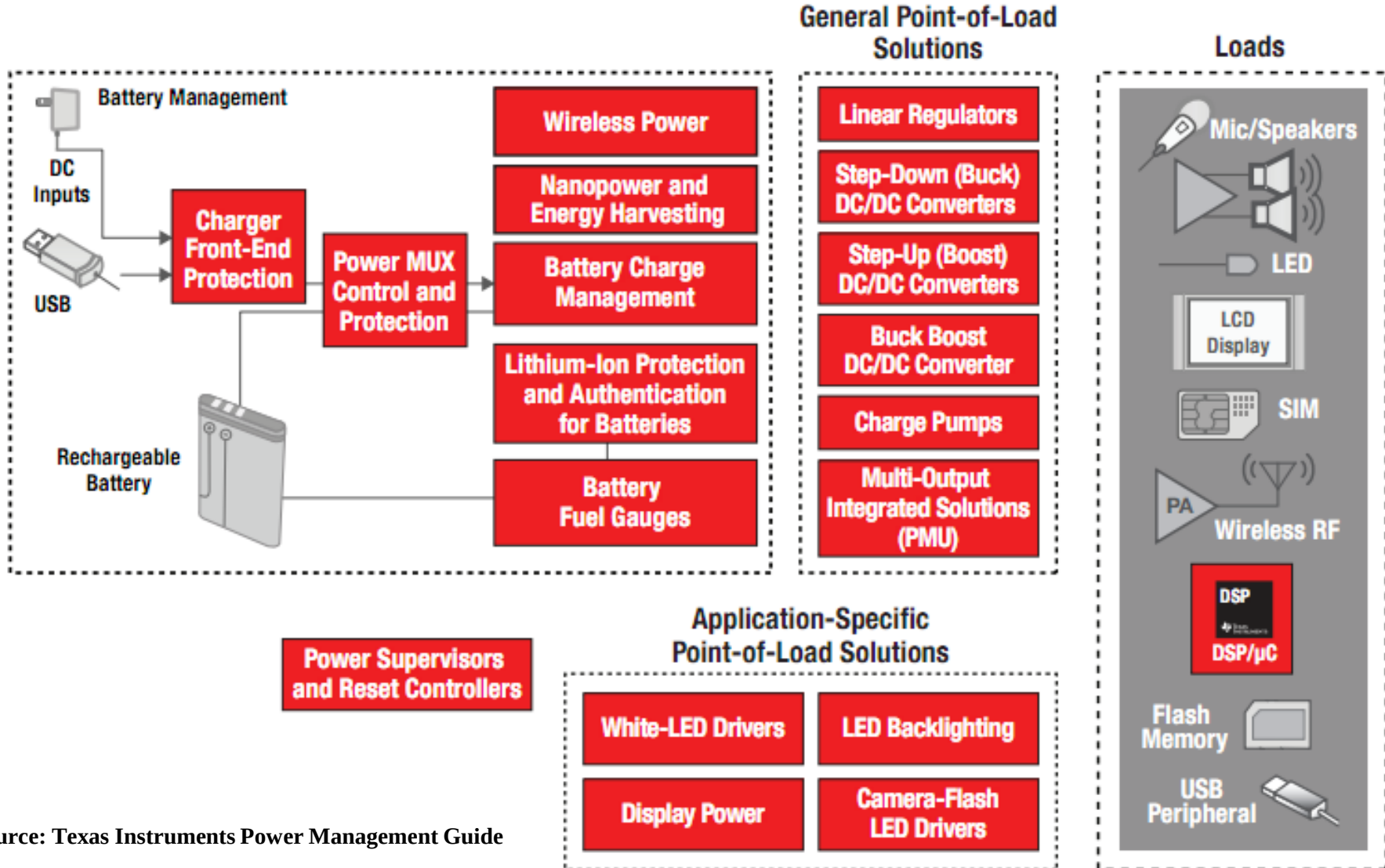
Source: Based on or includes content supplied by Informa Tech (former IHS Markit Technology), "Power Semiconductor Market Share Database – 2018", September 2019

2015 share	Change
12%	3.1%
11%	0.7%
6%	-0.1%
5%	0.1%
4%	0.1%
4%	0.0%
4%	0.4%
4%	-0.2%
3%	-0.7%
3%	0.0%
N/A	N/A
43%	-0.5%
\$34,011.4 m	-2.6%

and separately in 2014 and

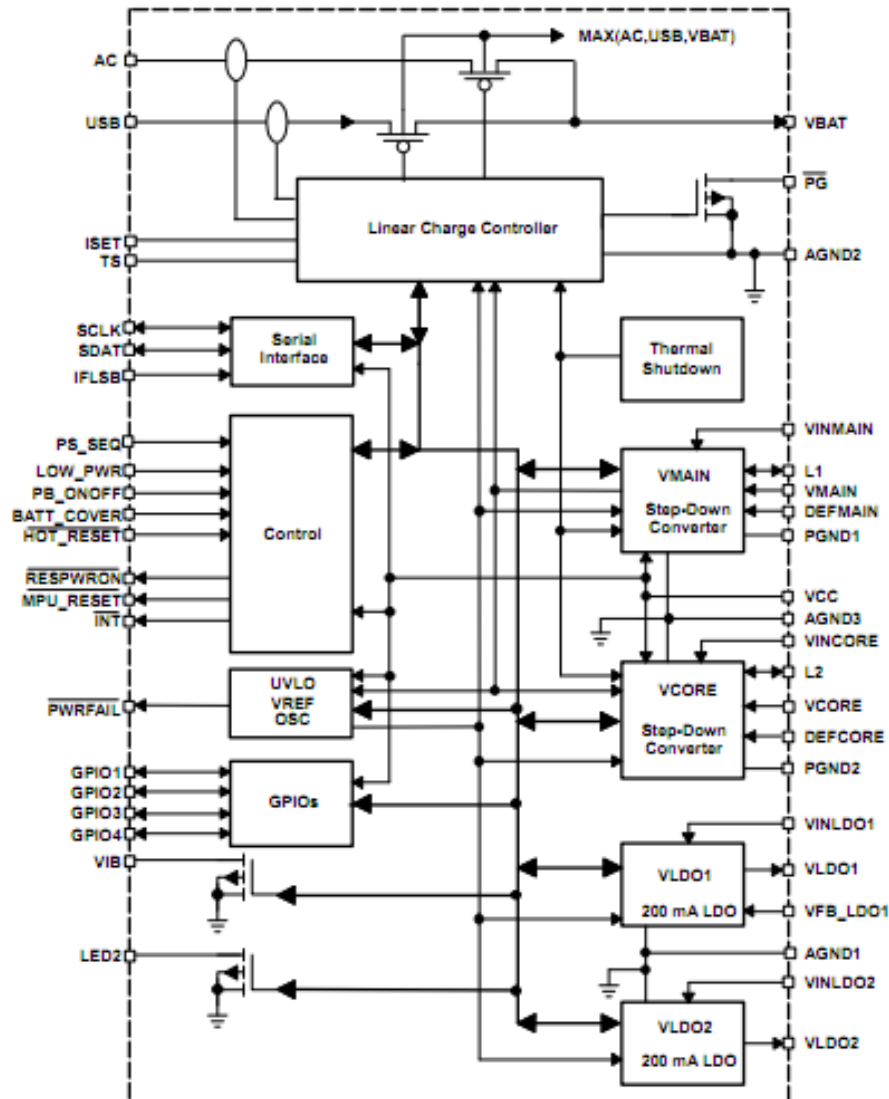
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PMU Building Blocks



Source: Texas Instruments Power Management Guide

A Typical PMU Architecture



TI TPS65010

- Linear charger for single Li-ION cell
- Dual inputs from USB or from wall plug 100 mA / 500 mA
- 1 A 95% efficient step-down converter
- 400-mA 90% efficient step-down converter
- 2x 200-mA LDOs
- 70- μ A quiescent current
- 1% reference voltage

System Composition

- **References**

- Voltage, current, time, etc.
- Temperature-, supply-, process-independent

- **Regulators**

- Regulate voltage, current, and/or time
- Against load, supply, and transient variations
- Stable and temperature- and process-independent.

- **Chargers**

- Manage and charge a battery to its optimum point.

- **Power Mixers**

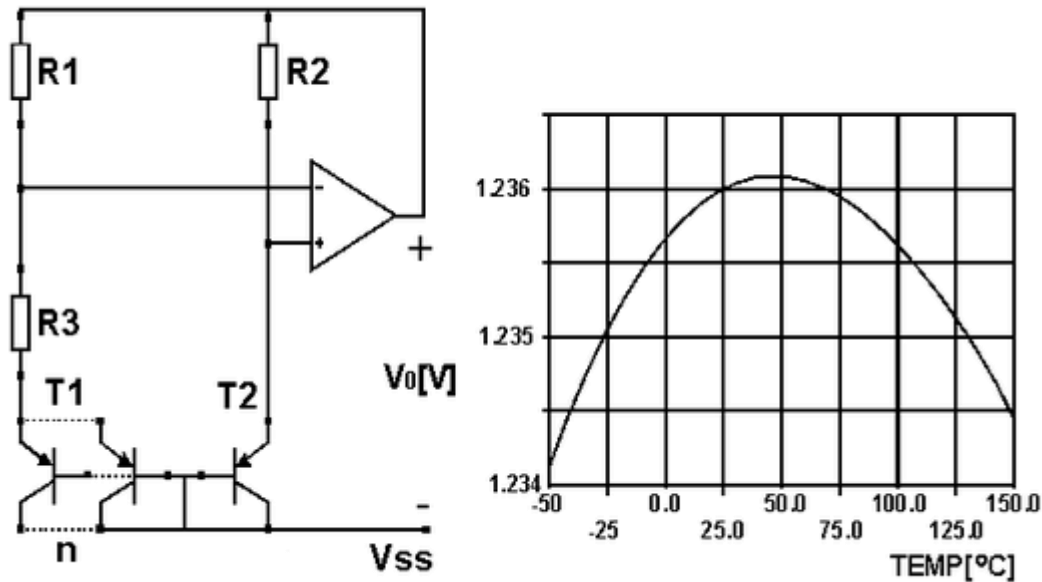
- Manage and mix power from sources.

- **Supporting Circuits**

- Over-current protection, thermal protection, etc.



Bandgap Reference



- Provides a temperature and supply-independent voltage reference.
- Required for comparison in other PMU building blocks to stabilize the voltages in a negative feedback loop.
- For proper operation of the PMU, BGR should have **minimal dependence on load and line, highest accuracy and lowest temperature coefficient.**

Regulator Types

- **Linear Regulators**

- Regulates V_{out} by linearly and continuously (analog) modulating the conductance of a series switch.

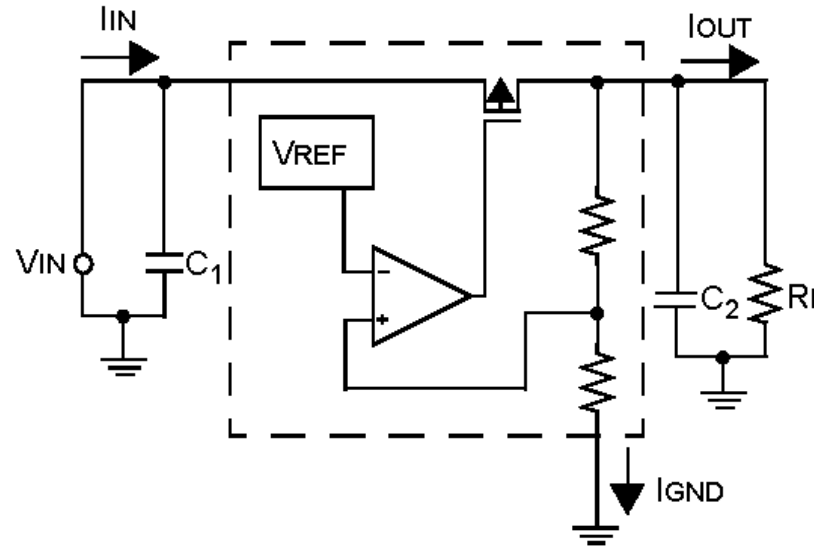
- **Switched Regulators**

- Regulates V_{out} by transferring energy from storage to load in alternate cycles (analog + digital)

Linear	Switched
Limited Output Range ($V_{out} < V_{in}$)	Flexible ($V_{out} \leq$ or $\geq V_{in}$)
Low Cost (PCB, Si, etc.)	Expensive (PCB, Si, etc.)
Low Noise	Switching Noise
Fast	Slow
Low Efficiency ($\eta \leq V_{out}/V_{in}$)	High Efficiency ($\eta \approx 80 - 95\%$)
Low-power, Low-noise Apps	High-power Apps



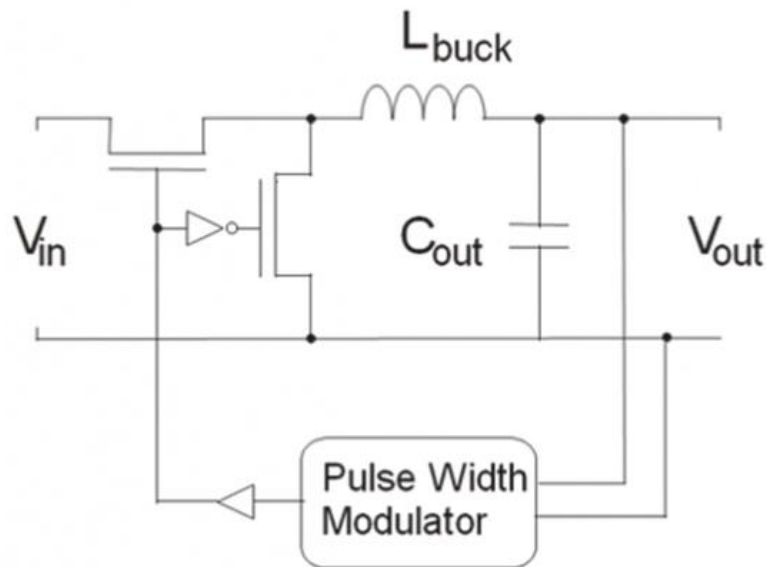
Low Drop-Out Regulators (LDOs)



$$\eta = \frac{V_{out}}{V_{in}}$$

- Provides ripple-free supply voltage.
- Should be **stable for various load conditions**.
- A huge pass transistor requires careful layout.
- Specifications include **PSR, line regulation, load regulation, and accuracy**.
- LDOs for digital loads should have **fast transient response**.

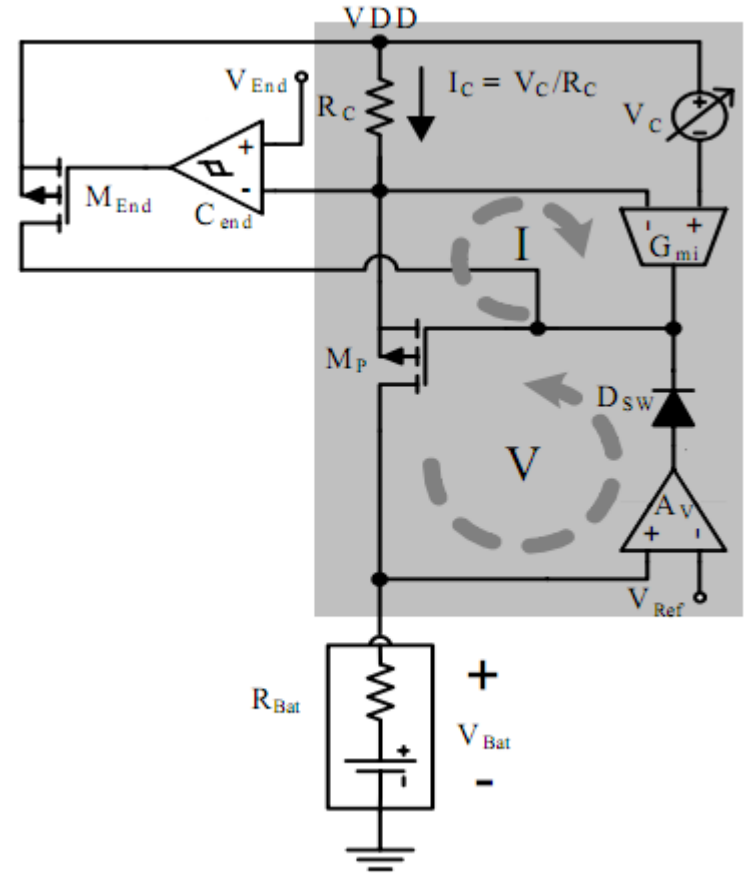
Buck Step-Down DC-DC Converter



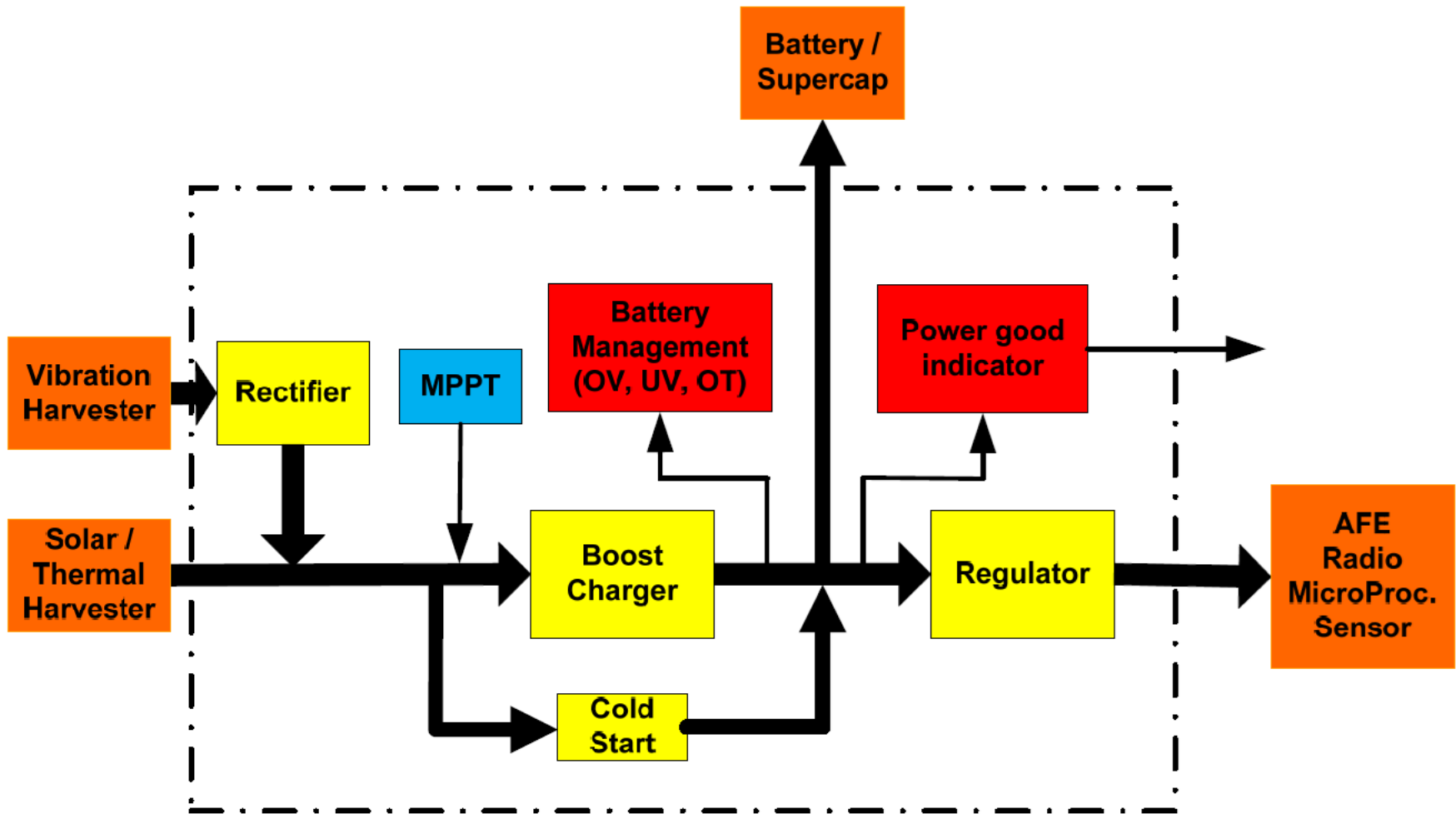
- Provides high-efficiency step-down voltage conversion.
- Should be **stable for various load conditions**.
- The PWM requires a clock generator and voltage comparison.
- Specifications include **ripples and ripple frequency, PSR, line regulation, load regulation, and accuracy**.

Battery Charging Controller

- Multiple loops for constant voltage and constant current charging
- Power-path control functionality
- Specifications include **efficiency, accuracy, and smooth transitions.**



Energy Harvesters



Energy harvesters are power sources

[Y. Ramadass, TI]



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- Building Blocks of a PMU
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 - LDOs
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 - Battery Charging Controller
 - Energy Harvesters
- **Power Management Glossary**



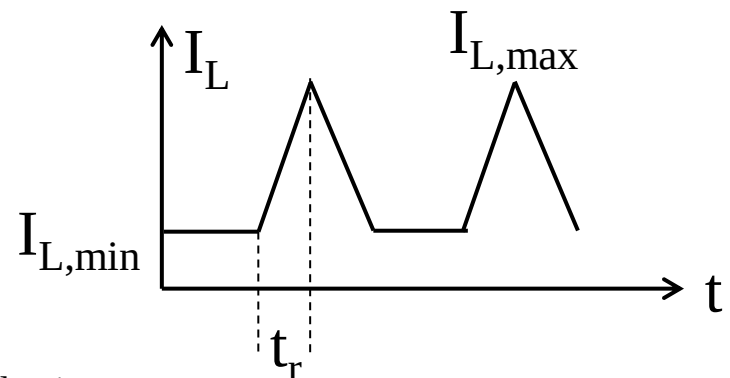
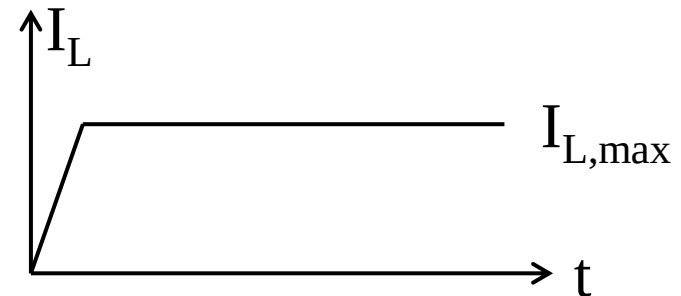
Load Types

- **According to load current**

- Zero load $\rightarrow I_L = 0$
- Light load $\rightarrow I_L = \text{microAmps to few milliAmps}$
- Heavy load $\rightarrow I_L = \text{hundreds of milliAmps}$

- **According to variations**

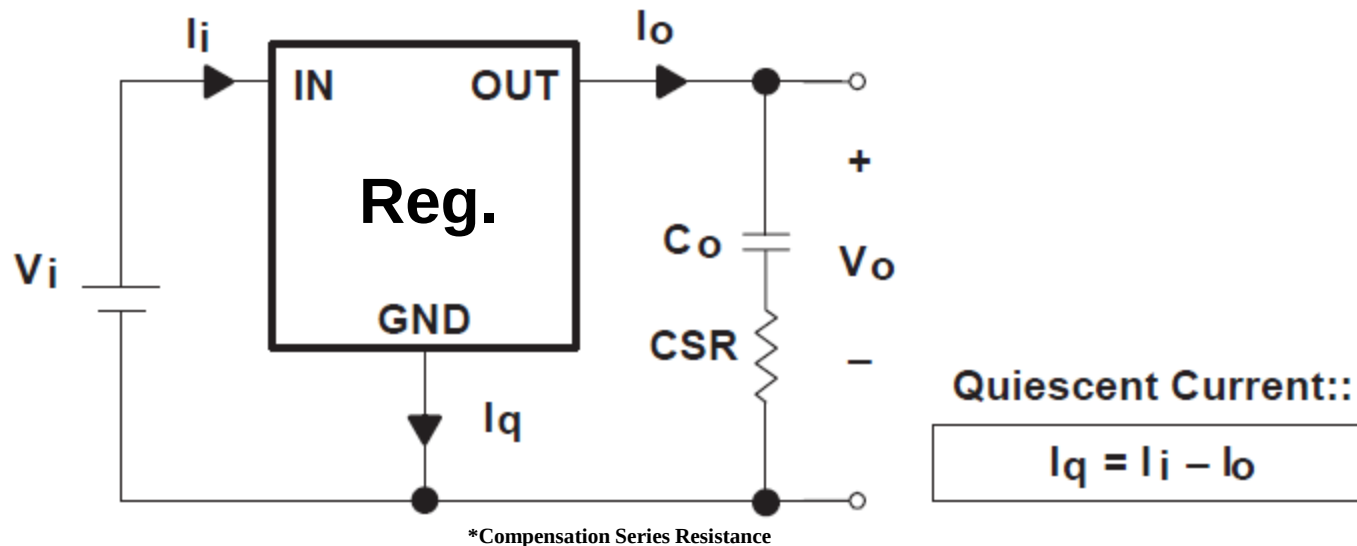
- Static or analog loads
- Dynamic or digital loads



Regulator Specifications (1)

- **Quiescent Current (I_q)**

- The average current consumed in the regulator not in the load

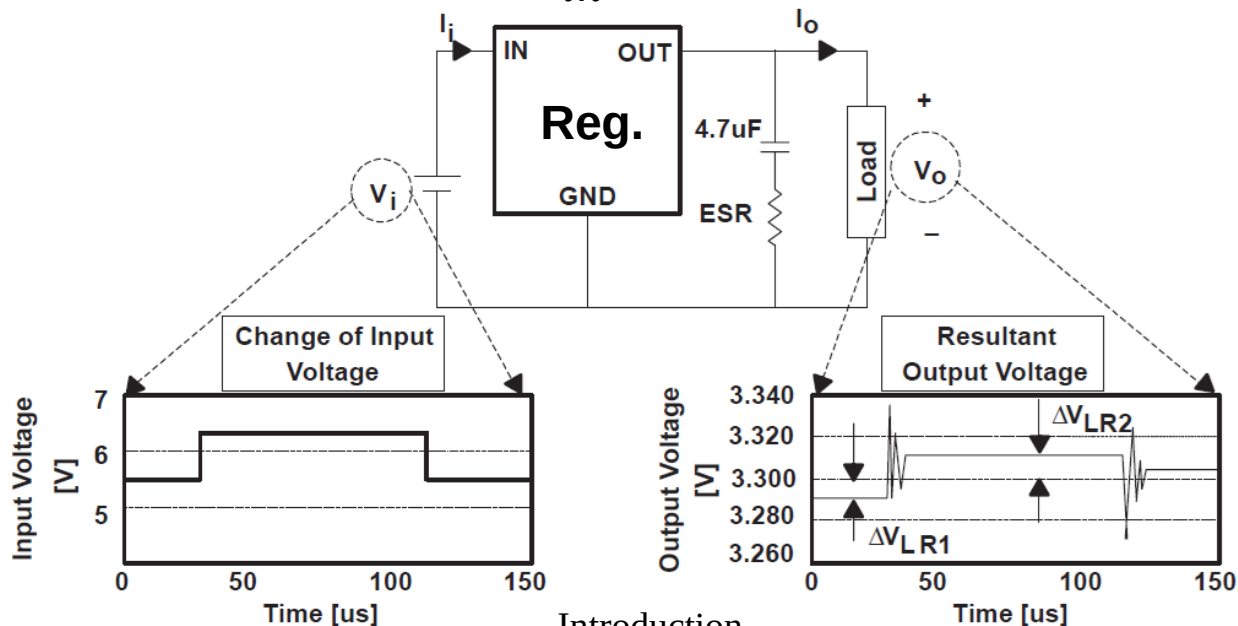


- **Efficiency (η)**

- Power output / Power input = $\frac{V_{out}I_L}{V_{in}I_{in}} = \frac{V_{out}I_L}{V_{in}(I_L + I_q)} \times 100$

Regulator Specifications (2)

- **Standby/Shutdown Current (I_{sd})**
 - The current consumed in the regulator when disabled
- **Line Regulation**
 - A measure of the circuit's ability to maintain the specified output voltage with varying input voltage
 - $Line\ Regulation = \frac{\Delta V_{out}}{\Delta V_{in}}$

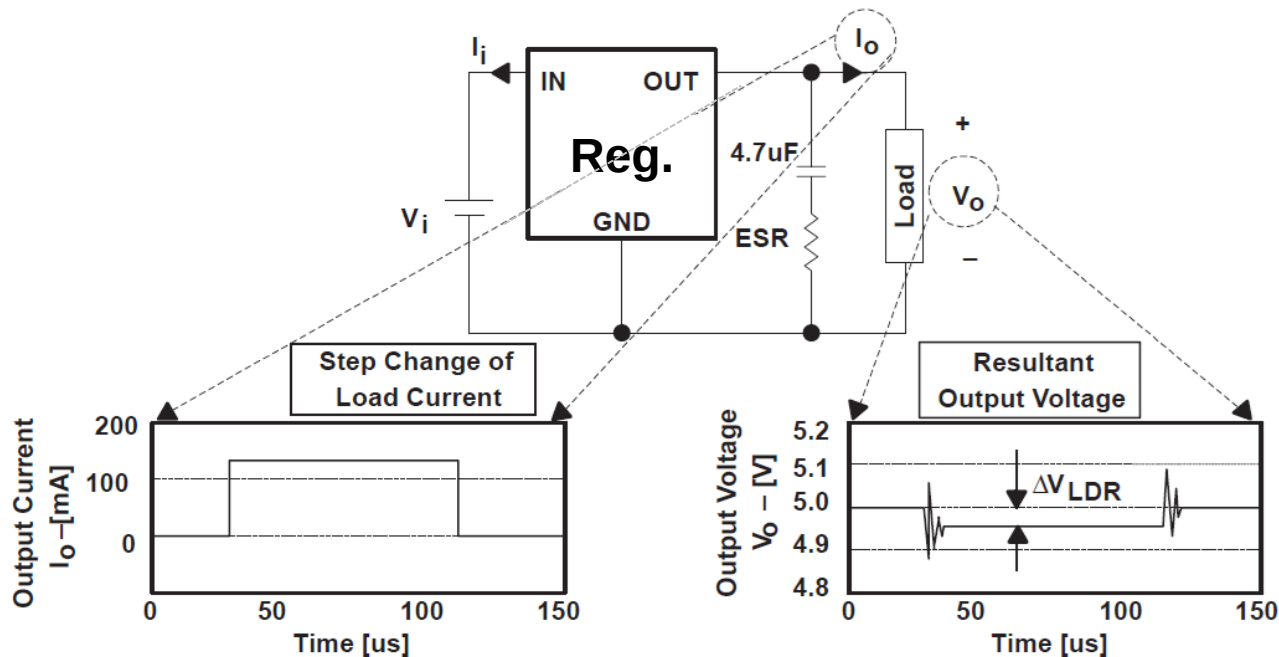


Regulator Specifications (3)

- **Load Regulation**

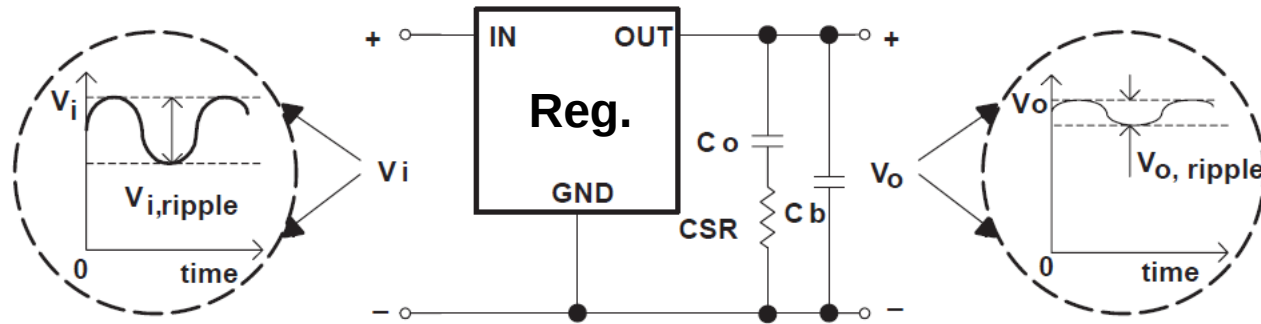
- A measure of the circuit's ability to maintain the specified output voltage under varying load conditions

- $Load\ Regulation = \frac{\Delta V_{out}}{\Delta I_L}$



Regulator Specifications (4)

- **Power Supply Rejection (PSR)**
 - Measures the regulator's ability to prevent the regulated output voltage fluctuating caused by input voltage variations.
 - The same relation for line regulation applies to PSR except that the whole frequency spectrum is considered.



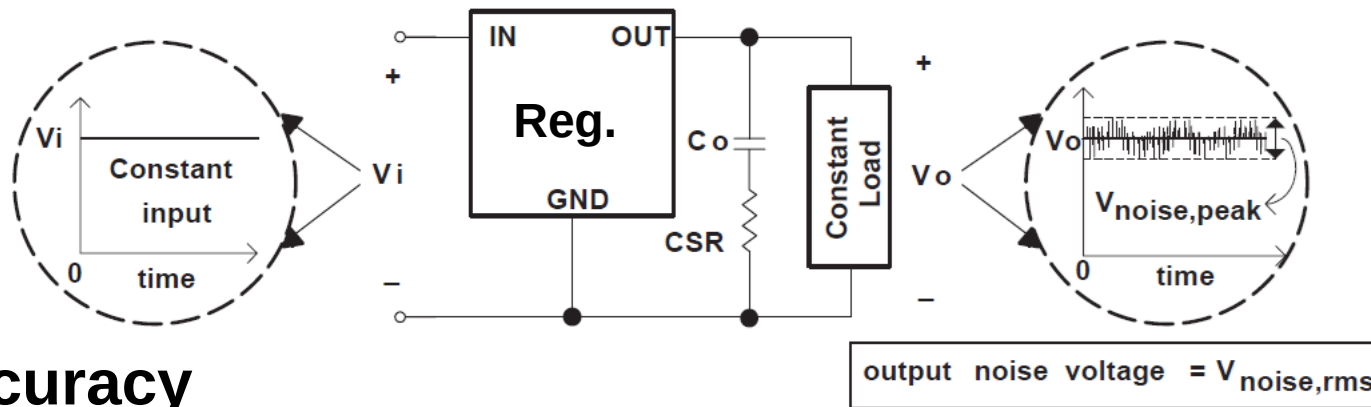
Ripple Rejection:

$$PSR = \frac{V_{o, ripple}}{V_{i, ripple}} \text{ at all frequencies}$$

Regulator Specifications (5)

- **Output Noise Voltage**

- The RMS output noise voltage over a given range of frequencies under the conditions of a constant output current and a ripple-free input voltage.



- **Accuracy**

- Difference between actual output and desired output in percentage = $\frac{V_{out} - V_{out,desired}}{V_{out,desired}} \times 100$

Course Contents

- **Bandgap Reference**
- **Linear Regulators**
- **Inductor-Based Switching Regulators**
- **Inductor-less Switching Regulators**
- **Battery Chargers**
- **Energy Harvesting**
- **Auxiliary Circuits**

